

Little Red Dots: Super-Eddington Stellar Envelopes, Not Overmassive Black Holes

Type IIIn supernovae and eta Carinae's Great Eruption offer a unified model for JWST's enigmatic Little Red Dots.

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Background

JWST's Little Red Dots show blackbody-like continua, broad Balmer lines, and lack variability. Similar features appear in Type IIIn supernovae and eta Carinae's Great Eruption.

Are LRDs overmassive black holes or something else? Their unique combination of traits defies standard AGN or galaxy explanations.

What this paper is selling

The authors propose that LRDs are intermediate-mass engines enshrouded in dense winds, mimicking stellar eruptions. This wind physics could explain the entire LRD life cycle, from enshrouded youth to black hole seeds.

Key results to remember

- LRDs, Type IIIn SNe, and eta Carinae share blackbody continua and H α profiles.
- Engine mass < 10⁵ solar masses, challenging overmassive black hole claims.
- Super-Eddington ($L_{\text{bol}}/L_{\text{edd}} > 5$) intermediate-mass systems power the emission.

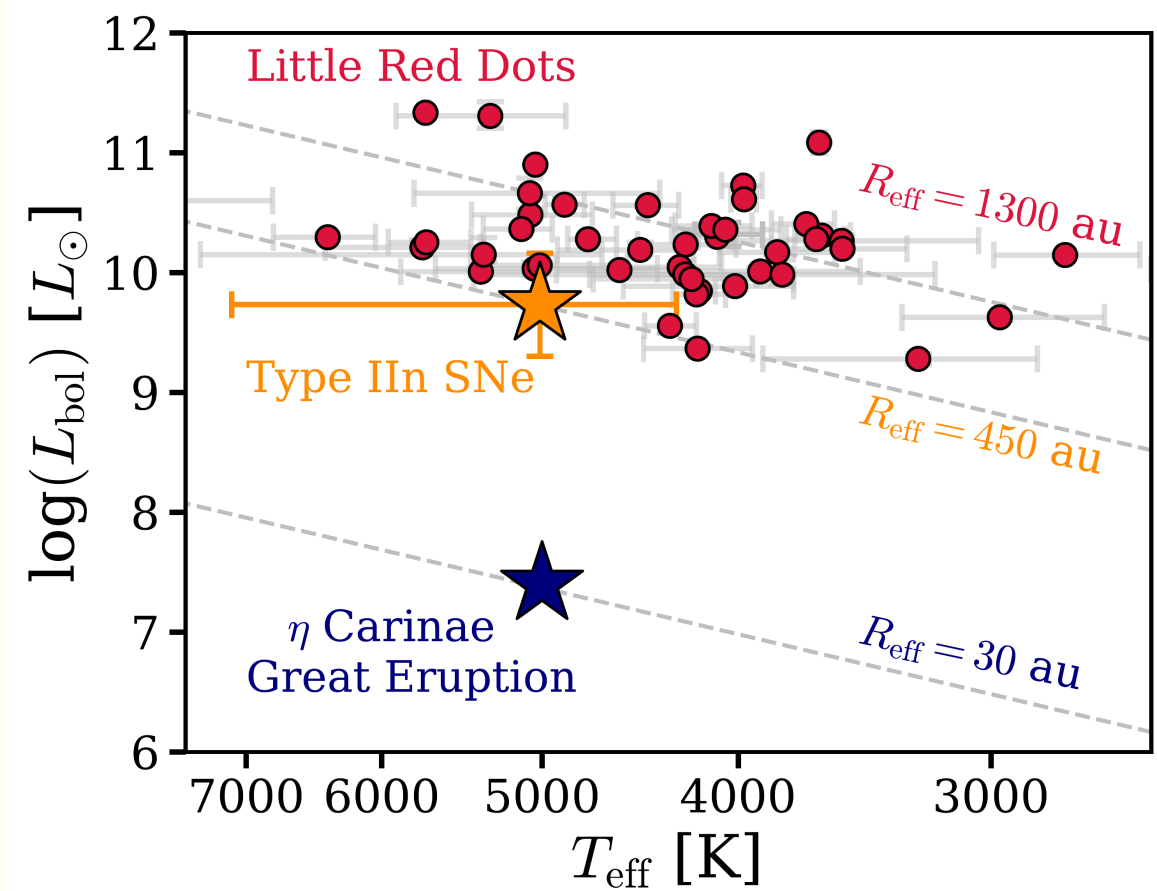


Fig. 4. LRDs, Type IIIn SNe, and eta Carinae cluster at similar effective temperatures, spanning four decades in luminosity.

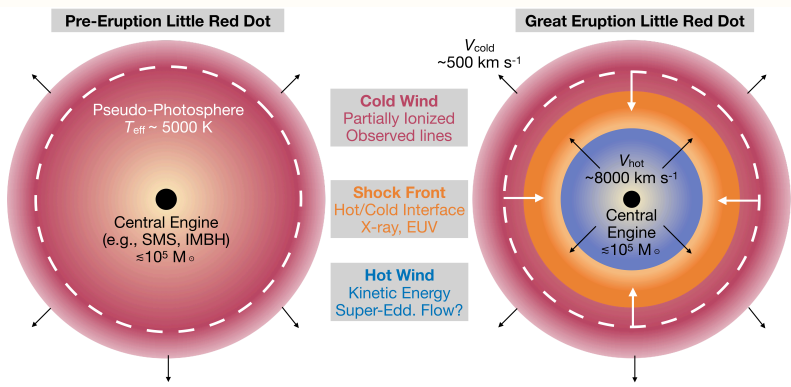


Fig. 8. Schematic: a cold wind creates a pseudo-photosphere; hot winds shock, converting kinetic energy into radiation.